



Best Technologies for Sustainable Environmental Solutions

PlanetDISK®

RBC – Rotating Biological Contactor Technology
Biological Wastewater Treatment Units
QUOTATION

Global Tech LLC

70 m³/day CAPACITY PlanetDISK® STP

OFFER INFORMATION	
Offer Number	YDD R0 07 08 2026 1 MX 1 70 25 114
Flow Rate	70 m ³ /day
Influent BOD ₅	350 mg/l
Effluent BOD ₅	40 mg/l
Wastewater Temperature	19 °C

August 7, 2026



john.doe@globaltech.com

August 7, 2026

Dear John Doe (CTO) ,

We thank you for your Wastewater Treatment Projects quotation request. The following are the details of what we can do for this specific project and how we can serve best for your environmental needs as effectively, economical and sustainable as possible.

Rotating Biological Contactor (RBC) is highly effective, innovative, and advanced method of domestic and industrial wastewater treatment with several unique advantages among other technologies where you can find the details inside our offer.

One of the advantages of the Rotating Biological Contactor technology is that the fixed film accumulated on the disks will tolerate sudden and long changes in hydraulic and organic loads. In addition, power consumption, compared to the conventional method is up to 80% less and the maintenance does not require constant attention of skilled person or sophisticated instrumentation.

PlanetDISK® Rotating Biological Contactor is the registered trademark of internationally recognized RBC equipment manufactured by PlanetTEK Environmental and Treatment Technologies Inc. since 2004. CE Certified **PlanetDISK®** RBC technology is manufactured in Türkiye with German know-how and sold world-wide!

PlanetTEK is the registered and the only qualified RBC equipment supplier for the Turkish state tenders for villages and state WWTP projects.

R&D for **PlanetDISK®** RBC Technology process has been established through extensive pilot plant testing as well as on-field real life operating experience since 2004. We are proud to successfully cooperate on various projects across the world with many well-established environmental and water treatment companies. We have many projects on 5 continents, near 40 countries and a qualified RBC equipment supplier for **Coca-Cola, UNDP, NATO MILITARY BASE IN PRISHTINA KOSOVA, VEOLIA, ARAMCO, METITO OF UAE, JSW STEEL-TEXAS-USA, HEAL GROUP OF AUSTRALIA, SOLIS A.E. OF GREECE, XELERA OF MEXICO, THYSSEN KRUPP, UNILEVER, INTERNATIONAL ROTARY CLUBS, CAPITAL WATER OF CHILE and KfW of Germany, AZERSU OF AZERBAIJAN! Projects** It has been 20 years since the REVOLUTION began!

PlanetDISK® RBC Technology has been tested under some of the toughest conditions such as at BTC Pipeline Stations of **BOTAŞ – BP- TEKFEN where the ambient temperature is less than -20°C as well as at 60°C in Doha-Qatar**. At any condition, **PlanetDISK®** RBC Technology yielded excellent results, both mechanically and process-wise.

We are very proud of our meaningful impact on Planet Earth. We are expanding internationally, and our goal is to share our success with your esteemed company. We would like to be the solution partners in this project in your part of the Planet where water is scarce and always the most precious.

I sincerely hope that we present a tangible, sustainable and valuable solution for this particular project of yours. Should you have any questions, please contact us and let us communicate. Put us at work and let us be at your service, at World's service.

Yours truly,

Hürriyet Necdet AYDOĞAN

Mech. Eng. And MSc. Indust. Eng.

Founder / CEO

ABOUT US

PlanetTEK Environmental and Treatment Technologies Inc. Since 2004; It produces the world's most widespread German wastewater treatment technology, **PlanetDISK®** Biological Wastewater Treatment systems, as 100% Turkish-made, in its Istanbul Gebze factory. From NATO Military Base Kosovo to ARAMCO Saudi Arabia, from Mexico to Seychelles, from Greece to Batumi, from AZERSU Azerbaijan to Casablanca, from Pakistan to the other end of the world, Australia, From Botswana to USA. **PlanetTEK** Environmental and Treatment Technologies Inc., which purifies the wastewater of tens of thousands of people on the continent and reuses some of the treated water; It has **ISO 9001: 2015 Quality Management System, ISO 14001: 2015 Environmental Management System** certificates in order to register the quality of “ **PlanetTEK** ”, by focusing on its standards within the framework of the necessity of good quality, continuity, innovative structure and technological progress.



Index

1. Design Parameters.....	5
2. Capital Expenditure (CAPEX).....	6
3. Scope of Supply.....	7
3.1. PlanetTEK Environment and Treatment Technology Inc. Scope of Supply.....	7
3.2. Client Scope of Works.....	7
4. Payment Conditions.....	8
5. Conditions of Assessment.....	8
6. Delivery Time (Excluding Shipment).....	8
7. Operation Cost of PlanetDISK® and Comparison with Other Technologies.....	9
7.1. Operation Expenditure (OPEX).....	9
7.1.1. Detailed Energy Operation Cost.....	9
7.1.2. Consumables and Maintenance Cost.....	9
7.2. Energy Saving Against Other Technologies.....	9
7.3. Carbon Footprint of Operation Comparison with Other Technologies.....	10
7.4. Economical Saving Against Other Technologies Including CAPEX and OPEX (for 10 years).....	11
7.5. Amortization of the System When Water Recycling Included.....	11
8. Design of Treatment Plant.....	12
9. Technical Details.....	13
10. Mechanical Equipment.....	14
11. Piping Works (If Included).....	15
12. Electrical Works (If Included).....	15
13. Warranties.....	16
13.1. Mechanical Warranty.....	16

1. Design Parameters

Domestic or industrial Wastewater Treatment Plant was designed according to influent and effluent concentrations which are shown in Table 1.

Table 1. Wastewater Influent and Effluent Concentration Limits of Wastewater Treatment Plant

Parameter	unit	Influent (*)	Effluent (!)
Design Flow Rate (Q)	m ³ /d	70	-
Daily BOD ₅ Load	kg/d	24.50	-
Plant Design Flow Rate	m ³ /h	2.92	-
Biochemical Oxygen Demand (BOD ₅)	mg/L	350	40
Chemical Oxygen Demand (COD)	mg/L	630	72
Total Suspended Solids (TSS)	mg/L	350	<40
Oil and Grease	mg/L	≤25	<20
pH	-	6 – 9	6 – 9
Wastewater Temperature	°C	15 - 32	15 - 32
Accepted Temperature	°C	19	-

(*) Wastewater should not contain any chemical which is harmful for the bacteria inside wastewater. Rainwater and swimming pool water should not be allowed into the wastewater treatment system. Grease and oil should not be allowed into the system. Wastewater from kitchens should pass through oil traps in kitchens.

(*) No animal manure should be allowed into the system.

(!) *Effluent Wastewater Concentrations are shown in Table 1.*

2. Capital Expenditure (CAPEX)

CAPEX						
Global Tech LLC				Offer Number : 0346		
Offer Reference Number : YDD R0 07 08 2026 1 MX 1 70 25 114						
No	Description	Piece	Unit Price	Total Price	Discount rate	Total Price after Discount
1.	MECHANICAL EQUIPMENT					
1.1.	Physical Treatment Units (Primary Treatment)					
1.1.1.	Coarse and Fine Screen Options					
1.1.1.1.	Manually Cleaned Coarse Screen	1	912.00 €	912.00 €	%5	866.40 €
1.1.1.2.	Manually Cleaned Fine Screen	1	958.00 €	958.00 €	%5	910.10 €
1.1.2.	Grit-Oil Trap Plates	4	126.00 €	504.00 €	%5	478.80 €
1.1.3.	Feeding Pump (1 duty + 1 standby) - City Pumps Ranger 10 35	2	542.30 €	1,084.60 €	%5	1,030.37 €
1.2.	Biological Treatment Units (Secondary Treatment)					
1.2.1.	PlanetDISK® MX 1 RBC Unit; - Epoxy Painted AISI 1045 (C45) Carbon Steel Solid Shaft, - Wet Parts SS304 Stainless and Galvanized Coated Steel, - 114 disk / unit per Shaft, - Disk Surface Area 752.40 m² / unit	1	32,036.00 €	32,036.00 €	%25	24,027.00 €
1.2.2.	LS_15 Lamella Separator Final Clarifier Tank	1	7,224.00 €	7,224.00 €	%25	5,418.00 €
1.2.3.	smt 100A Final Clarifier Tank Sludge Pump	1	395.85 €	395.85 €	%5	376.06 €
2.	CIVIL WORKS					
2.0.1.	Screen and Grit-Oil Trap Channel					
2.0.2.	First Primary Sedimentation Tank					
2.0.3.	Second Primary Sedimentation Tank					
2.0.4.	Equalization Tank					
3.	INSTALLATION MATERIAL					
3.0.1.	All piping and electrical installation works	1	3,817.00 €	3,817.00 €	%5	3,626.15 €
4.	ELECTRICAL WORKS					
4.0.1.	PlanetDISK® Control Panel	1	6,121.00 €	6,121.00 €	%5	5,814.95 €
5.	TRANSPORTATION					
5.0.1.	40' HC Container	1	13,027.00 €	Optional	%0	Optional
5.0.2.	Truck	1	0.00 €	Optional	%0	Optional
TOTAL				53,052.45 €	Total Discount	10,504.62 €
*Optional and Supply Locally equipment are not included in the total price.				Grand Total Discounted Price		42,547.83 €
* V.A.T. , All Other Expenses and Taxes are EXCLUDED						
* EXWORKS PlanetTEK Kocaeli Gebze Factory						
* Validity : 30 days						
Supply locally : equipment should be obtained locally for better warranty conditions.						

*Our offer price is limited to the equipment given in the CAPEX list. Additional requests from the customer will be priced separately in the detailed project.

3. Scope of Supply

3.1. PlanetTEK Environment and Treatment Technology Inc. Scope of Supply

- Detailed engineering
- Detailed drawings (Flow Diagram, P&I, General Layout, etc.)
- Supply of all Mechanical Equipment mentioned in Section 2.
- Electrical Control Panel (If included)
- Supervision for Installation of the system (All costs belongs to the Client. Flight tickets, accommodation, visa, local costs etc.)
- Supervision for Commissioning of the system (will be discussed later)
- Operation and Maintenance Instruction Manual in English with 2 copies and 1 USB
 - o Manuals,
 - o Operating instructions,
 - o Service instructions,
 - o Trouble shooting,
 - o Safety instructions,
 - o Spare parts list,
 - o Component catalogues.

3.2. Client Scope of Works

- All Civil Works
- To supply electricity to control panel of treatment plant,
- To supply wastewater to the treatment plant,
- To supply utility water to the site area of treatment plant,
- To discharge treated water or irrigation water,
- The final ground arrangement,
- Concrete platform to place the units,
- The units should be isolated against the effect of high or low temperature. If it will not be so cold, then you can cover only the top of all units something like a roof. But if it is cold, it is better to cover all sides of all units like placing all units inside of a building.
- Customs clearance, and related taxes, and inland transportation,
- All expenses and procedure that will occur to get the necessary discharge permit from local authorities,
- Laying and manufacturing of underground pipes and concrete transition parts according to the given project (Handrails, platforms, all stainless steel, HDPE pipes and piping etc.)
- Arrangement of the projects given by Planettek according to the desired standards and norms,
- Crane and other equipment needed to mobilize and place the RBC units on site,
- Supply of chemical and biological materials at start up (mainly chlorine etc.)
- The treated wastewater should be discharged to a receiving environment such as a lake or river under the limits of regulations.
- Visa, Flight Tickets, Accommodation and Local Food Expenses of Supervisor
- Shipment organization and cost depending on INCOTERM
- Generator and isolation (Generator has to be used for validity of warranty terms of the system.)

4. Payment Conditions

The pricing quoted in this proposal is based on the following terms of payment;

- 50 % Down-payment at Purchase Order/Contract (Manufacturing of equipment will start upon receipt of advance payment)
- 50 % before container loading at **PlanetTEK** manufacturing premises. The payment will be made before loading in containers upon approval by Client.
All payments are via bank transfer. All Bank and financial fees belong to Client if occurring in client's country. PlanetTEK is responsible for all bank and financial fees occurring in Türkiye.

PlanetTEK shall only proceed with procurement and production of equipment and materials upon receipt of a formal purchase order/contract and advance payment.

Alternative mode of payment against additional fees

30% down-payment with order and 70% payment via L/C, or 100% payment via L/C.

L/C: payment secured by irrevocable letter of credit of an international accepted bank, with delivery by PlanetTEK (cost of delivery to be quoted for and added separately). 100% payment at sight of freight papers, with 3% additional surcharge cost, minimum € 2.000, on L/C part (not on down-payment).

5. Conditions of Assessment

- 1.The characteristics and the quantity of the wastewater has to be in accordance with the assumed figures given to us are chosen by us, which are shown in the design calculation.
2. In case additional equipment for the wastewater treatment would be necessary because of not foreseeable substances it is not included in this quotation.
3. PlanetTEK guarantees for the values given in the table "design calculation". Other limits are not guaranteed. The wastewater must not contain toxic or inhibiting substances. Standard sanitary conditions of inlet wastewater are assumed unless otherwise specified.
4. Equivalent technical modifications are reserved.
5. Individual people or countries with trade restrictions / Embargo – individual assessment at time of order (non-binding quote for these cases).
6. In case of non-pick-up of an already manufactured unit, a grace period of maximum 6 weeks is acceptable. After this grace period a storage fee of 0,5% per started month of whole project value will be charged until pick-up. After 26 weeks the unit will be subject to auctioning / change of ownership with 50% devaluation or more.

6. Delivery Time (Excluding Shipment)

Our offer is prepared according to the following work schedule. If our offer is confirmed, work schedule will be revised and submitted to you because of negotiations.

Preparation of layouts and process projects	~1	Week
Manufacturing and supply of STP equipment	~8-10	Weeks
Piping and Installation works	~3-4	Weeks
Commissioning and Training	~1	Week

7. Operation Cost of PlanetDISK® and Comparison with Other Technologies

7.1. Operation Expenditure (OPEX)

OPERATION EXPENDITURE - OPEX	
GRAND TOTAL	

7.1.1. Detailed Energy Operation Cost

ENERGY OPERATION COST						
Total Hydraulic Load	70.00 m ³ /hour	Unit Installed Power	Total Installed Power	Power Consumed	Daily Working	Electricity Consumption
Energy Price for 1 kW.hour	13.00 € cent					
MECHANICAL EQUIPMENTS	Unit	kW	kW	%	hour	kW.hour/day
Physical Treatment Units (Primary Treatment)						
Equalization Tank Feed Pump	1	0.80	0.80	90%	24	17.28
Biological Treatment Units (Secondary Treatment)						
PlanetDISK® MX RBC Unit	1	0.37	0.37	90%	24	7.99
Secondary Clarifier Sludge Pump	1	0.80	0.80	90%	1	0.72
TOTAL ELECTRICITY CONSUMPTION						25.99
Electricity consumption per 1 m ³ wastewater					0.37	kW.hour / m ³
Electricity consumption per 1 m ³ wastewater					4.83	cent / m ³
Electricity Consumption Cost					3.38	€ / day
					1,233	€ / year
*** "Energy Price for kw/hour = 13 € cent" is the rate in TURKEY & given for comparison purposes						

7.1.2. Consumables and Maintenance Cost

CONSUMABLES and MAINTENANCE COST				
Description	Total Amount	Consumption	Unit Price	Total Price
CONSUMABLES AND MAINTENANCE EXPENSES				
Biological Treatment Units (Secondary Treatment)				
PlanetDISK® MX DBD Unit				
Bearing Grease	2 bearing	0.50 kg/year.bearing	6 €/kg	6.00 €/year
Gearbox Oil	1 gearbox	7.40 lt/year.gearbox	4 €/lt	29.60 €/year
GRAND TOTAL				35.60 €/year

7.2. Energy Saving Against Other Technologies

ENERGY COMPARISON of PlanetDISK® RBC vs ACTIVATED SLUDGE SYSTEM				
PlanetDISK® Unit		Activated Sludge		
Motor Reduction Gear			Blower	Sludge Feed Pump
MX 1 Unit Number	1 pieces	Number	1 pieces	1 piece
MX 1 Motor Power	0.37 kW	Power	2.00 kW/hour	2.20 kW/hour
Total Power	0.37 kW/hour	Total Power	2.00 kW/hour	2.20 kW/hour
Power Consumption	90%	Power Consumption	90%	90%
Total Actual Power to be used	0.33 kW	Total Actual Power to be used	1.80 kW	1.98 kW
Electricity Price	0.13 €/kW	Electricity Price	0.13 €/kW	0.13 €/kW
Daily Working Time	24 hour/day	Daily Working Time	24 hour/day	4 hour/day
Yearly Working Time	365 day/year	Yearly Working Time	365 day/year	365 day/year
PlanetDISK® Unit Energy Consumption	379 € /year	Activated Sludge Energy Consumption	2,050 € /year	376 € /year
Comparing PlanetDISK® Unit with other systems yearly, electric power saving is 2,046 € /year				
For 10 years electric power saving price is equal to 20,464 € /10 year				
and together with blower and diffusers maintenance cost, it will be approx 21,000 €				

7.3. Carbon Footprint of Operation Comparison with Other Technologies

CARBON FOOTPRINT					
Electricity Consumption Definitions	Unit		Electricity Consumption Definitions	Unit	
RBC System Energy Consumption	8.0	kw/day	AS System Energy Consumption	51.1	kw/day
RBC System Energy Consumption	2,917	kW/year	AS System Energy Consumption	18,659	kW/year
CO2 Emission Coefficient	0.43	kg/eMWh	CO2 Emission Coefficient	0.43	kg/eMWh
Annual Carbon Footprint	1.3	ton/year	Annual Carbon Footprint	8.0	ton/year
Due to electricity consumption, there is an annual debt of	57	trees to nature.	Due to electricity consumption, there is an annual debt of	365	trees to nature.

7.4. Economical Saving Against Other Technologies Including CAPEX and OPEX (for 10 years)

PlanetDISK® DBD TECHNOLOGY vs ACTIVATED SLUDGE SYSTEM 10 YEARS TOTAL COST OF OWNERSHIP (TCO) ANALYSIS				
	Initial Investment Cost (Excl. Civil Works)	Yearly Energy Cost	Yearly Operator Cost	Yearly Maintenance & Spare Parts Cost
PlanetDISK®	42547.829999999999	379.2204	1219	350
Aktif Çamur	31910.8725	2425.644	4876	1050
10 Years Total Savings with PlanetDISK®: 70,000.00 €				

7.5. Amortization of the System When Water Recycling Included

AMORTIZATION TABLE						
IF THE IRRIGATION WATER IS SUPPLIED FROM MUNICIPAL WATER		Daily water required	Monthly water required	Yearly water required	Unit Price of Municipal Water	Yearly Total Water Use Cost
UNIT		m ³ /day	m ³ /month	m ³ /year	€	€/year
VALUE		70.00	2,100	14,700	1.5900	23,373
IF THE IRRIGATION WATER IS SUPPLIED FROM WWTP	WWTP CAPEX		Amortization Time			
UNIT	€		Year			
VALUE	42,548		1.82			
In 22 Months, the WWTP is amortizing itself.						
*** "Unit Price of Municipal Water = 1.5900 €/m ³ " is given for comparison purposes.						

8. Design of Treatment Plant

Global Tech LLC

70 m³/day (People Equivalent) Capacity - Discharge to Nature Rotating Biological Contactor (RBC) Sewage Treatment Plant (STP) Offer

PROJECT INFO

Hydraulic Load	: 70 m ³ /day
Organic Load	: 24.50 kg/day (70 m ³ /day x 350 mg/l)
Elimination of BOD in Pre Treatment	: 8.09 kg/day (33%)
Treated Organic Load in PlanetDISK® RBC Unit	: 16.42 kg/day
Wastewater Temperature	: Min 15°C-Max 32°C
Accepted Temperature	: 19 °C
Wastewater Retention Time in PlanetDISK® RBC Unit	: 1.54 hours (>45 minutes minimum)

*These parameters are directly given by client. / chosen according to our experiences and literature reviews.

Wastewater Source

Only domestic wastewater from toilets, sinks, shower, dishwash and laundry.
Wastewater should not contain any chemicals which is harmful for the bacteria inside wastewater.
Rainwater and swimming pool water which may contain chlorine and chemicals should not be allowed into the ST system. Fat, Oil and Grease (FOG) should not be allowed into the system.
Wastewater from kitchens definitely should pass through oil traps in kitchens.

Proposed System

- 1 PlanetDISK® MX 1 units - 114 disks of 2.05 m diameter disks, GRP (fiber glass) housing.
- TOTAL 114 disks x 6.6 m² x 1 units = 752.4 m² surface area.
- 1 LS 15 Lamella Separator final sedimentation tank, GRP (fiberglass) body.
- TOTAL 15 m² x 1 units = 15 m² lamella surface area.

Disk Surface Area Calculation

$$\pi \times r \times r \times 2 \text{ sides} \times 114 \text{ disks/unit} \rightarrow 3.14 \times 1.025 \times 1.025 \times 2 \times 114 = 752.2 \text{ m}^2/\text{unit}$$

9. Technical Details

Quotation No	2026 / 0346		
Quotation Reference No	YDD R0 07 08 2026 1 MX 1 70 25 114		
Client	Global Tech LLC		
Design Parameters			
· Wastewater Source	Only domestic wastewater from toilets, sinks, shower, dishwash and laundry.		
· Final Destination of Treated Water	Discharge to Nature		
· Hydraulic Load	70	m ³ /day	
· Hourly Flow Rate	2.92	m ³ /hour	
· Peak Flow Rate	5.83	m ³ /hour	
· Organic Load (BOD)	25	kg/day	
· Wastewater Temperature	19	°C	
· Wastewater Retention Time in PlanetDISK® RBC Unit	1.54	hours	
· Number of PlanetDISK® MX 1 RBC Units	1	units	
· Number of Disks	114	disks/unit	
· Disk Diameter	2.05	m	
· Surface Area of 1 Disk	6.6	m ² /disk	
· Surface Area of PlanetDISK® MX 1 RBC Unit	752.4	m ² /unit	
· Total Disk Surface Area in this Project	752.4	m ²	
· Final Sedimentation Tank with Lamella Separator LS 15	1	units	
· 1 Unit Lamella Separator Final Sedimentation Tank LS 15	15	m ² /unit	
· Total Lamella Surface Area in Final Sedimentation Tank LS 15	15	m ²	
Offer content			
✓	1	set	Manually Cleaned Coarse Screen
✓	1	set	Manually Cleaned Fine Screen
✓	4	units	Sand-Oil Trap Plates
✓	2	units	Equalization Tank Feed Pumps
✓	1	units	PlanetDISK® MX 1 RBC Unit
✓	1	units	PlanetDISK® MX 1 RBC Unit Cover (Fiberglass)
✓	1	units	LS 15 Lamella Separator Final Sedimentation Tank
✓	1	units	LS 15 Lamella Separator Final Sedimentation Tank Sludge Pump
			Civil & Construction Works – by Client
✓	1	set	Piping & Electrical Works
✓	1	unit	PlanetDISK® Control Panel
✓	-	-	Supervision, Engineering, Commissioning & Training

10. Mechanical Equipment

MECHANICAL EQUIPMENTS	
Manually Cleaned Coarse Screen	
Quantity	1 Unit
Bar Spacing	30 mm
Material	Galvanized ST37 Carbon Steel
Type	Bar or perforated type screen
Manually Cleaned Fine Screen	
Quantity	1 Unit
Bar Spacing	10 mm
Material	Galvanized ST37 Carbon Steel
Type	Bar or perforated type screen
Grit-Grease Trap Baffles	
Quantity	4 Units
Material	Fiberglass (GRP) or Polypropylene or Composite
Dimension	1000 x 1000 mm
Equalization Tank Feed Pumps	
Quantity	1 Duty + 1 Standby
Capacity	2.9 m ³ /h @ 5.9 mSS
Motor	0.8 kW
Type	Submersible
Brand	City Pumps Ranger 10 35
Material	Cast Iron Body
PlanetDISK® MX RBC Unit - With Lid	
Quantity	1 Unit(s)
Disk Diameter	2.05 m
Motor Power	1 x 0.37 kW
Surface Area per Disc	6.6 m ² / disc
Number of Discs per Unit	114 discs / unit
Minimum Total Disc Area	752.4 m ²
Rotation Speed	3 – 4 rpm
Unit Dimensions [L x W x H]	3,350 x 2,370 x 2,650 mm
Discs	PP (Polypropylene), Virgin Material, Single Piece Disc, Thickness: 1.5 mm - 2 mm
Solid Shaft	Epoxy Painted AISI 1045 (C45) Carbon Steel, diameter: 85 mm
Body	Fiberglass (GRP)
Chassis	Powder Coated Carbon Steel
Wetted Parts	AISI 304 Stainless Steel

MECHANICAL EQUIPMENTS	
<i>RBC Unit Manufacturer</i>	PlanetTEK Environmental and Water Wastewater Treatment Technologies Inc.
<i>Motor Manufacturer</i>	WAT or equivalent
<i>Gearbox Manufacturer</i>	PGR or equivalent
Blower	
<i>Quantity</i>	1 Unit
<i>Capacity</i>	15 m ³ /h @ 200 mbar
LS 15 Lamella Separator Final Sedimentation Tank	
<i>Quantity</i>	1 Unit(s)
<i>Manufacturing Material</i>	GRP Body & Powder Coated Carbon Steel Chassis
<i>Lamella Plate Area</i>	15 m ² /tank
<i>Tank Volume</i>	2 m ³ /tank
<i>Manufacturer</i>	PlanetTEK Inc.
<i>Cleaning</i>	The sludge accumulated at the bottom is automatically discharged via the sludge pump. When necessary, lamella surfaces are manually washed with pressurized water.
Final Clarifier Tank Sludge Pump	
<i>Quantity</i>	1 Unit(s)
<i>Manufacturer</i>	Sumak or Equivalent
<i>Capacity</i>	10 m ³ /hour @ 8.1 mSS
<i>Motor</i>	0.8 kW
<i>Type</i>	Centrifugal Type
CIVIL WORKS	
Screen Channel	
<i>Quantity</i>	1 Unit
<i>Capacity</i>	3 m ³ wet volume
<i>Material</i>	Reinforced Concrete
<i>Retention Time</i>	30 minutes at peak flow rate of 5.8 m ³ /h
Primary Clarifier Tank	
<i>Quantity</i>	1 Unit
<i>Capacity</i>	15 m ³ wet volume
<i>Material</i>	Reinforced Concrete
<i>Retention Time</i>	2.5 hours at peak flow rate of 5.8 m ³ /h
Secondary Clarifier Tank	
<i>Quantity</i>	1 Unit
<i>Capacity</i>	12 m ³ wet volume

MECHANICAL EQUIPMENTS	
<i>Material</i>	Reinforced Concrete
<i>Retention Time</i>	2 hours at peak flow rate of 5.8 m ³ /h
Equalization Tank	
<i>Quantity</i>	1 Unit
<i>Capacity</i>	12 m ³ wet volume
<i>Material</i>	Reinforced Concrete
<i>Retention Time</i>	2 hours at peak flow rate of 5.8 m ³ /h

*The size, capacity and quantity of all mechanical equipment and construction works may vary in the final project.

11. Piping Works (If Included)

Pipes and fittings to be used in the Wastewater Treatment Plant will have TSE Quality certificate. Pipelines will be selected from U-PVC materials.

12. Electrical Works (If Included)

Electrical and control panels have a mimic diagram and thermic magnetic automatic contactor switches. The control panel will be plastic. There will be warning alarm lights and a sound system to warn against rotor rotation failure.

There will be a mimic diagram on the panel. System's flow diagram and lead that show equipment's on / off / fault positions can be seen on the mimic diagram. So, all the equipment that works or not, can be seen easily by any operator.

Control panel instruments will have TSE Quality certificate. All the instruments/equipment inside the electrical control panel are selected from well-known brands such as ABB.

Electrical Works	
Quantity	1
Manufacturer	PlanetTEK
Protection Class	IP 65
Voltage	380 V / 50 Hz - Three-Phase
Specifications	ABB, Siemens, Schneider or eq. contactor etc. materials, thermic magnetic automatic switches, mimic diagram with red-green led lights on the panel, sound alarm system, emergency button

*** It is highly recommended to use a generator and voltage regulator in order to prevent any possible drawbacks due to potential power cuts.**

13. Warranties

13.1. Mechanical Warranty

Heart of the system is the **PlanetDISK® RBC equipment** and carries **two (2) years** warranty against all defects and due to materials or workmanship from the date of shipment if machinery is operated and stored in accordance with PlanetTEK design specifications and/or instructions.

All other mechanical parts (**Screens, lamella separator, grease and grit trap plates**) carry **2 years warranty**.

All electrical parts including **electrical motors, reduction gear and pumps** carry a **2-years warranty** from the manufacturers.

Bearings at both sides of the shaft carry **2 years warranty** as long as lubricated with the specified grease type regularly, at least once a month.

All the warranty conditions start after commissioning or latest after 6 months after the shipment, whichever comes first.

PlanetDISK® Units must be placed on a level surface until in use.

When the wastewater treatment plant is not in use, **PlanetDISK®** Units must be turned on for at least 30 minutes/month. If there is no electricity at site, a power generator will be used.

Should the Client fail to comply with these rules, warranty conditions will not apply for any potential mechanical failures.

Only treated/plated steel, virgin, vacuum-formed single piece PP disks and special certified plastic materials are used in the construction of **PlanetDISK®** compact units.

It is vitally important that the rotor of RBC equipment rotates continuously. Interruptions can be at most 30-40 minutes. If there is any probability of more interruptions, the system must be connected to an automatic start power generator. Longer and frequent interruptions of the rotation will cause **PlanetDISK®** unit to fail mechanically and cause serious damage to shaft, gear and other moving parts of the Equipment. PlanetTEK Inc does not give a guarantee to **PlanetDISK®** RBC units against such frequent or long power failures.

Failing to remove the grease, fat, or oil from the grease trap and WW with FOG over the given criteria limits will both mechanically damage the system and biologically harm the process.

Should there be a need for technical service to serve the system for any mechanical failures, PlanetTEK Service man will come to service for free of service charge if the failure is covered by warranty. In any case, the customer will pay for travel and lodging expenses.

We thank you again for your trust and interest in our company. We are looking forward to being your partner in your wastewater treatment solutions for a better Planet, our only home.

"One day everything will be cheaper than water. ®"

TEAM PlanetTEK

